Differential Equations and Slope Fields

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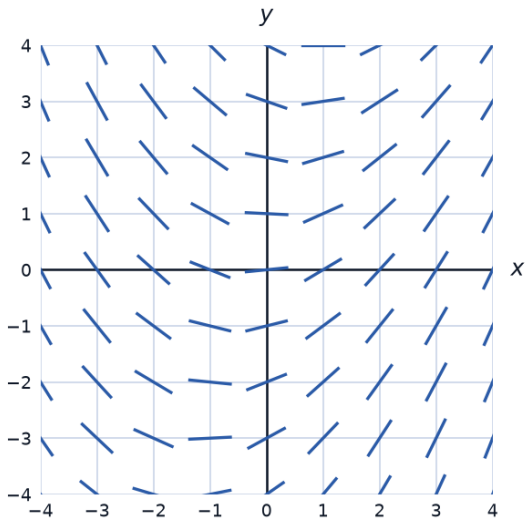
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Verifying solutions to differential equations

Verify that $y = Ce^{3x}$ is a solution to the differential equation $\frac{dy}{dx} = 3y$ for any constant C.

2 [Foundational] • Target: Verifying solutions to differential equations

Verify that $y = x^2 + C$ is a general solution to $\frac{dy}{dx} = 2x$, and find the particular solution satisfying $y(1) = 5$.



3 [Foundational] • Target: Solving separable differential equations

Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 3$.

4 [Foundational] • Target: Slope fields

For the differential equation $\frac{dy}{dx} = x - y$, find the slope at the points $(0, 0)$, $(1, 0)$, and $(1, 1)$, as would be drawn on a slope field.

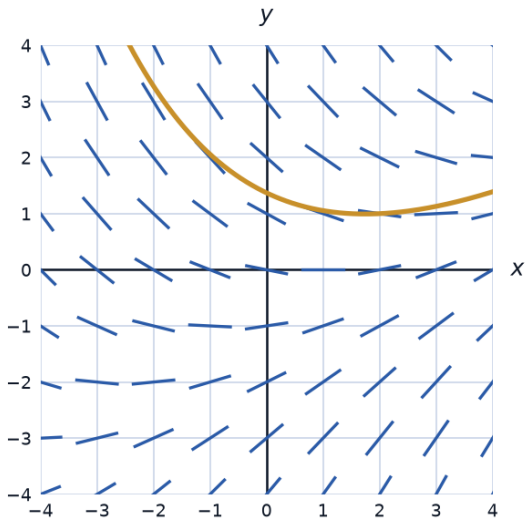
5 [Foundational] • Target: Slope fields

Explain how a slope field for $\frac{dy}{dx} = ky$ (exponential growth/decay) would look different for $k > 0$ versus $k < 0$.

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6 [Foundational] • Target: Exponential growth and decay models

A bacteria culture grows according to $\frac{dP}{dt} = 0.2P$, with $P(0) = 500$. Find $P(t)$ and the population after 10 hours.



7 [Foundational] • Target: Exponential growth and decay models

A radioactive substance decays according to $\frac{dA}{dt} = -0.05A$. If $A(0) = 100$ grams, find how long it takes to decay to 50 grams (the half-life).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

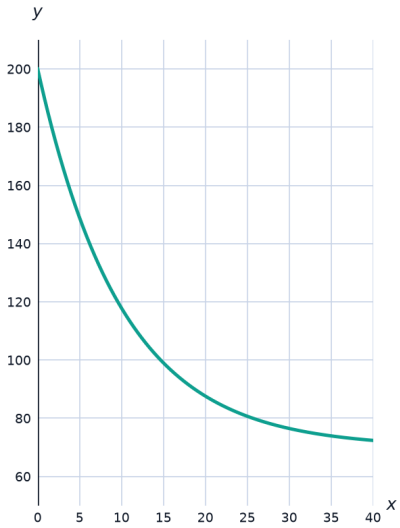
8 [Intermediate] • Target: Matching slope fields to differential equations

A slope field shows horizontal segments (slope 0) along the entire x-axis and along the entire y-axis, with slopes elsewhere resembling the product xy. Identify the differential equation this slope field represents.

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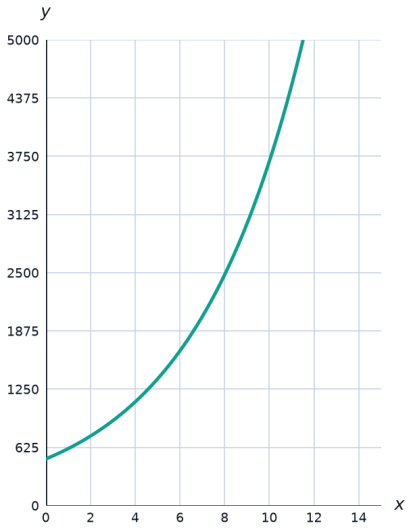
9 **[Intermediate]** • **Target: Newton's Law of Cooling**

An object cools according to $\frac{dT}{dt} = -k(T - 70)$, where $70°\text{F}$ is room temperature. If $T(0) = 200°\text{F}$ and $k = 0.1$, find $T(t)$.



10 **[Intermediate]** • **Target: Visualizing an exponential growth solution**

The graph shows the bacteria population $P(t) = 500e^{0.2t}$ from the earlier problem. Use the graph to estimate how long it takes for the population to double from its initial value of 500, then verify algebraically using $\ln 2$.



11 **[Intermediate]** • **Target: A logistic-style differential equation**

A population satisfies $\frac{dP}{dt} = 0.1P\left(1 - \frac{P}{200}\right)$. Find the equilibrium solutions (where $\frac{dP}{dt} = 0$), and explain which one is stable for a population starting near $P = 10$.

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12 **[Intermediate]** • **Target: A mixing problem modeled by a differential equation**

A tank contains 50 L of pure water. Brine with concentration 3 kg/L flows in at 2 L/min, and the well-mixed solution drains at the same rate. If $y(t)$ is the amount of salt (kg) at time t, the differential equation is $\frac{dy}{dt} = 6 - \frac{2y}{50} = 6 - \frac{y}{25}$. Find the equilibrium value of y, and explain what it represents physically.

13 **[Intermediate]** • **Target: Slope fields**

Level 1 · Foundations Find the particular solution to $\frac{dy}{dx} = \frac{3x^2}{y}$ subject to $y(0) = 4$.

14 **[Intermediate]** • **Target: Exponential growth and decay models**

Level 1 · FoundationsA population satisfies $\frac{dP}{dt} = 0.05P - 0.0001P^2$. Determine the carrying capacity L and evaluate $\lim_{t\to\infty} P(t)$ given initial condition $P(0) = 150$.

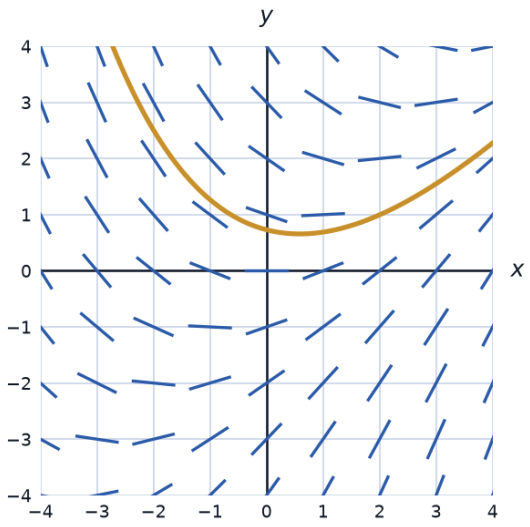
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Matching slope fields to differential equations** • **Cross-Lesson: 8.01 Area Between Curves**

[Differential Equations · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

16 **[Challenging]** • **Target: Newton's Law of Cooling** • **Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods**

[Unit 7BC · Logistic Models & Euler's Method · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.



17 **[Challenging]** • **Target: Visualizing an exponential growth solution** • **Cross-Lesson: 1.01 Limits Graphically and Numerically**

A weighted network has edges: DE = 1, BC = 2, AB = 3, CD = 4, BD = 5, CE = 6, AC = 7, AD = 8. Use Kruskal's algorithm to determine the minimum spanning tree and compute its total weight.

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18 [Challenging] • Target: A logistic-style differential equation • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Level 2 · MediumFind the slope of the tangent line $\frac{dy}{dx}$ to the polar cardioid $r = 1 - \cos\theta$ at $\theta = \frac{\pi}{2}$.

19 [Challenging] • Target: Additional separable equations • Cross-Lesson: 2.01 The Definition of the Derivative

A network has weighted adjacency matrix W. Explain how the minimum spanning tree algorithm (Kruskal's) differs from Dijkstra's algorithm.

20 [Challenging] • Target: A mixing problem modeled by a differential equation • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Level 1 · FoundationsConvert the Cartesian equation $x^2 + y^2 = 6y$ into polar form, and identify the resulting geometric shape.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Verifying solutions to differential equations

For the differential equation $\frac{dy}{dx} = -3x + 5y$, find the slope at the point $(-1,\ -4)$, as would be drawn on a slope field.

a. -19

b. -17

c. -15

d. 23

22 [Foundational] • Target: Solving separable differential equations

Solve the separable differential equation $\frac{dy}{dx} = \frac{x}{y}$, given $y(0) = 21$. Find $y(20)$ (the positive root).

a. 21

b. 29

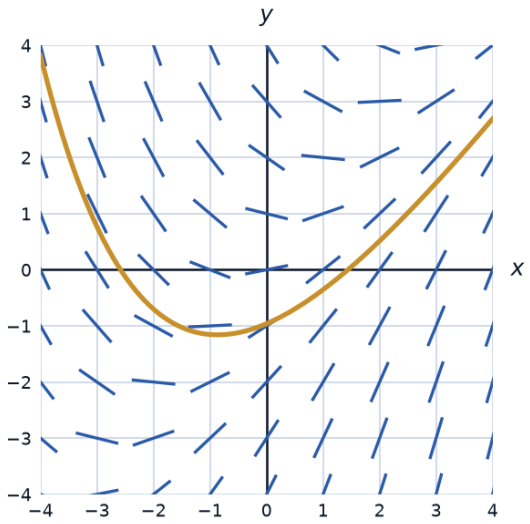
c. 20

d. 28

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23 [Foundational] • Target: Slope fields

A quantity grows according to $\frac{dP}{dt} = \frac{3}{20}P$, with $P(0) = 326$. Find $P(5)$, rounded to the nearest whole number.



a. 587

b. 594

c. 794

d. 690

24 [Foundational] • Target: Exponential growth and decay models

A quantity decays according to $\frac{dP}{dt} = -\frac{1}{4}P$, with $P(0) = 473$. Find $P(7)$, rounded to the nearest whole number.

a. 106

b. 95

c. 70

d. 82

25 [Foundational] • Target: Matching slope fields to differential equations

A substance decays according to $\frac{dA}{dt} = -\frac{1}{10}A$. If $A(0) = 126$ grams, find how long it takes to decay to 84 grams, to one decimal place.

a. 9.1

b. 2.8

c. 4.1

d. 5.3

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26 [Foundational] • Target: Newton's Law of Cooling

A tank's salt content $y(t)$ (kg) satisfies $\frac{dy}{dt} = 9 - \frac{y}{25}$. Find the equilibrium value of y.

a. 216

b. 34

c. 225

d. 250

27 [Foundational] • Target: Visualizing an exponential growth solution

A population of bacteria is modeled by the logistic differential equation $\frac{dP}{dt} = kP(1 - \frac{P}{L})$, where $P(t)$ is the population at time t in hours, k is the growth rate constant, and L is the carrying capacity. It is observed that $\frac{dP}{dt} = 200$ when $P = 100$, and the maximum growth rate occurs when $P = 500$. Find the carrying capacity L of this population.

a. 1000

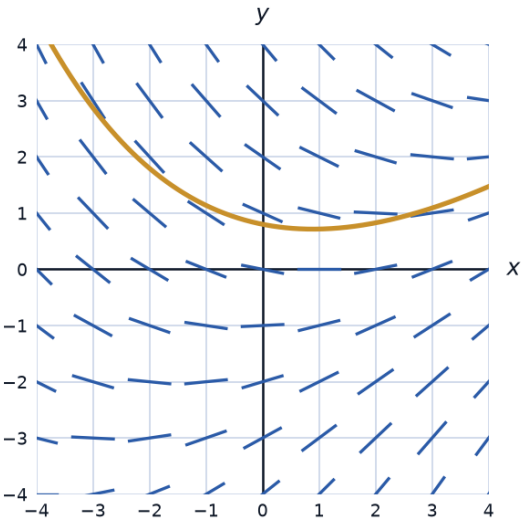
b. 750

c. 1200

d. 500

28 [Foundational] • Target: A logistic-style differential equation

The equation $y' = ky$ models exponential growth when:



a. $k = 0$

b. $k < 0$

c. $k = 1$

d. $k > 0$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Additional separable equations

Consider the logistic differential equation $\frac{dP}{dt} = kP(M - P)$, where $P(t)$ is the population at time t, k is the growth rate constant, and M is the carrying capacity. If $\frac{dP}{dt} = 0.05P(1000 - P)$, what is the carrying capacity of the population?

a. 1000

b. 0.05

c. 50

d. 0.00005

30 [Intermediate] • Target: A mixing problem modeled by a differential equation

Coffee at 70°C is placed in a 22°C room and cools to 45°C in 5 min (Newton's law of cooling). How long (min) until it reaches 35°C?

a. 7

b. $\frac{10\ln\left(\frac{48}{13}\right)}{\ln\left(\frac{48}{23}\right)}$

c. $\frac{5\ln(2)}{\ln\left(\frac{14}{9}\right)}$

d. $\frac{5\ln\left(\frac{48}{13}\right)}{\ln\left(\frac{48}{23}\right)}$

31 [Intermediate] • Target: Verifying solutions to differential equations

Newton's law of cooling: $\frac{dT}{dt} = k(T - T_s)$, $k < 0$. Steady state:

a. $T = 0$

b. $T = kT_s$

c. $T = T_s$

d. $T = T_0$

32 [Intermediate] • Target: Solving separable differential equations

Exponential decay: $\frac{dN}{dt} = -kN$, $k > 0$. Solution:

a. $N = N_0 e^{kt}$

b. $N = N_0/(1 + kt)$

c. $N = N_0 - kt$

d. $N = N_0 e^{-kt}$

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33 [Intermediate] • Target: Slope fields

Consider the differential equation $\frac{dy}{dx} = 2x$. Which of the following is a particular solution passing through the point $(1, 3)$?

a. $y = x^2 + 2$

b. $y = x^2 + 1$

c. $y = 2x + 1$

d. $y = 2x^2 - 1$

34 [Intermediate] • Target: Exponential growth and decay models

Logistic model: $\frac{dP}{dt} = rP(1 - P/K)$. At $P = K/2$, the growth rate is:

a. zero

b. minimum

c. maximum

d. negative

35 [Intermediate] • Target: Matching slope fields to differential equations

Carrying capacity K in the logistic model: as $P \to K$ from below, growth rate $dP/dt \to$

a. 0

b. K

c. ∞

d. -1

36 [Intermediate] • Target: Newton's Law of Cooling

Initial value problem: $y' = y$, $y(0) = 2$. Solution:

a. $y = e^{2x}$

b. $y = 2e^x$

c. $y = 2x + 2$

d. $y = xe^x$

37 [Intermediate] • Target: Visualizing an exponential growth solution

Separable ODE $\frac{dy}{dx} = xy$. Solution:

a. $y = x^2/2 + C$

b. $y = Ce^x$

c. $y = Ce^{x^2/2}$

d. $y = Ce^{x^2}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: A logistic-style differential equation • Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem

Separable ODE: $\frac{dy}{dx} = \frac{x}{y}$. Solution (implicit):

a. $x^2 + y^2 = C$

b. $y^2 = x^2$

c. $y = x + C$

d. $y^2 = x^2 + C$

39 [Challenging] • Target: Additional separable equations • Cross-Lesson: 2.01 The Definition of the Derivative

Logistic equation steady states (equilibria):

a. $P = 0$ only

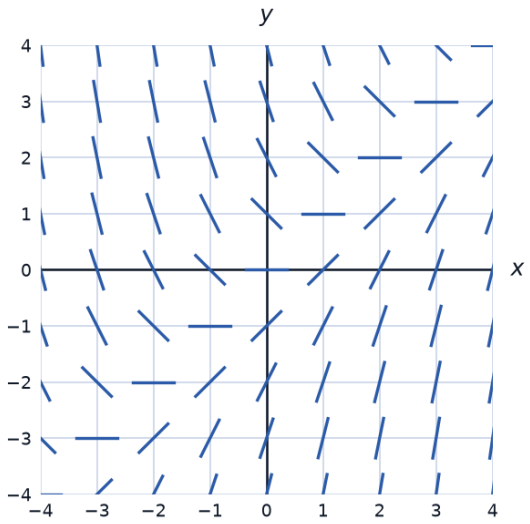
b. $P = K$ only

c. $P = 0$ and $P = K$

d. $P = K/2$

40 [Challenging] • Target: A mixing problem modeled by a differential equation • Cross-Lesson: 2.02 Product, Quotient, and Trigonometric Derivatives

Consider the slope field shown below. Which of the following differential equations could represent this slope field, assuming the slopes are defined at all points except where division by zero occurs? $\frac{dy}{dx} = \frac{x}{y}$



a. $\frac{dy}{dx} = \frac{x}{y}$

b. $\frac{dy}{dx} = \frac{y}{x}$

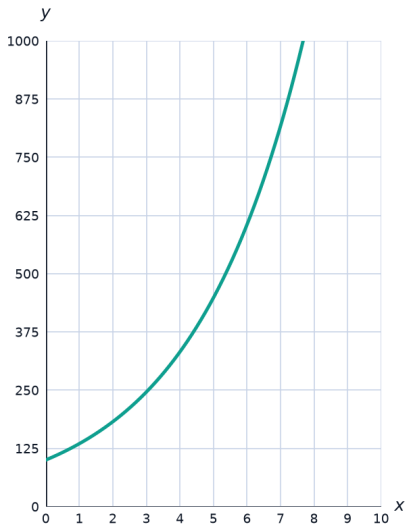
c. $\frac{dy}{dx} = x - y$

d. $\frac{dy}{dx} = x + y$

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41 [Challenging] • Target: Verifying solutions to differential equations • Cross-Lesson: 3.01 The Chain Rule

Logistic growth model is $\frac{dP}{dt} =$



a. kP

b. $kP(1 - P/M)$

c. $k(M - P)$

d. P^2

42 [Challenging] • Target: Solving separable differential equations • Cross-Lesson: 3.02 Implicit Differentiation and Inverse Functions

Consider a particle whose velocity vector $\mathbf{v}(t) = \langle x'(t), y'(t) \rangle$ satisfies the differential equation $y'' - 2y' + y = 0$ and the initial conditions $x(0) = 1, y(0) = 2, x'(0) = 0, y'(0) = 1$. Furthermore, the particle's path is such that the magnitude of its acceleration vector is always equal to the magnitude of its velocity vector, i.e., $||\mathbf{a}(t)|| = ||\mathbf{v}(t)||$. Find the value of $x(1)$.

a. $e^{-1} + 1$

b. $2e^{-1} + 1$

c. $e^{-1} - 1$

d. $2 - e^{-1}$

43 [Challenging] • Target: Slope fields • Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion

Picard's theorem guarantees existence and uniqueness of IVP solution when f and $\partial f/\partial y$ are:

a. constant

b. polynomial

c. continuous on a neighborhood of (x_0, y_0)

d. analytic everywhere

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44 [Challenging] • Target: Exponential growth and decay models • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

Newton's law of cooling: $T(t) = T_e + (T_0 - T_e)e^{-kt}$. If room temp $T_e = 20$, initial $T_0 = 80$, $k = 0.1$, $T(10) =$

a. $20 + 60$

b. $60e^{-1}$

c. $80e^{-1}$

d. $20 + 60e^{-1} \approx 42.1$

45 [Challenging] • Target: Matching slope fields to differential equations • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

A second-order ODE with repeated root $r = r_1$ has general solution:

a. $y = C_1e^{r_1x} + C_2e^{-r_1x}$

b. $y = C_1e^{r_1x}$

c. $y = (C_1 + C_2)e^{2r_1x}$

d. $y = (C_1 + C_2x)e^{r_1x}$

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